National
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# Physics
# Paper 1 — Relationships sheet

THURSDAY, 21 MAY
9:00 AM – 9:45 AM

*X8577622*

Qualifications Scotland
Teisteanasan Alba

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**Relationships required for Physics Higher**

$d = \bar{v}t$

$s = \bar{v}t$

$v = u + at$

$s = ut + \frac{1}{2}at^2$

$v^2 = u^2 + 2as$

$s = \frac{1}{2}(u + v)t$

$F = ma$

$W = mg$

$E_w = Fd$, or $W = Fd$

$E_p = mgh$

$E_k = \frac{1}{2}mv^2$

$P = \dfrac{E}{t}$

$p = mv$

$Ft = mv - mu$

$F = G\dfrac{m_1 m_2}{r^2}$

$t' = \dfrac{t}{\sqrt{1 - \left(\dfrac{v}{c}\right)^2}}$

$l' = l\sqrt{1 - \left(\dfrac{v}{c}\right)^2}$

$f_o = f_s\left(\dfrac{v}{v \pm v_s}\right)$

$z = \dfrac{\lambda_{observed} - \lambda_{rest}}{\lambda_{rest}}$

$z = \dfrac{v}{c}$

$v = H_0 d$

$W = QV$

$E = mc^2$

$I = \dfrac{P}{A}$

$I = \dfrac{k}{d^2}$

$I_1 d_1^2 = I_2 d_2^2$

$E = hf$

$E_k = hf - hf_0$

$v = f\lambda$

$E_2 - E_1 = hf$

$d \sin\theta = m\lambda$

$n = \dfrac{\sin\theta_1}{\sin\theta_2}$

$\dfrac{\sin\theta_1}{\sin\theta_2} = \dfrac{\lambda_1}{\lambda_2} = \dfrac{v_1}{v_2}$

$\sin\theta_c = \dfrac{1}{n}$

$V_{rms} = \dfrac{V_{peak}}{\sqrt{2}}$

$I_{rms} = \dfrac{I_{peak}}{\sqrt{2}}$

$T = \dfrac{1}{f}$

$V = IR$

$P = IV = I^2R = \dfrac{V^2}{R}$

$R_T = R_1 + R_2 + \ldots$

$\dfrac{1}{R_T} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \ldots$

$V_1 = \left(\dfrac{R_1}{R_1 + R_2}\right)V_S$

$\dfrac{V_1}{V_2} = \dfrac{R_1}{R_2}$

$E = V + Ir$

$C = \dfrac{Q}{V}$

$Q = It$

$E = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{1}{2}\dfrac{Q^2}{C}$

*path difference* $= m\lambda$ or $\left(m + \frac{1}{2}\right)\lambda$ where $m = 0, 1, 2\ldots$

$$\textit{random uncertainty} = \frac{\textit{max. value} - \textit{min. value}}{\textit{number of values}}$$

or

$\Delta R = \dfrac{R_{\max} - R_{\min}}{n}$

**Additional relationships**

**Circle**

circumference $= 2\pi r$

area $= \pi r^2$

**Sphere**

area $= 4\pi r^2$

volume $= \frac{4}{3}\pi r^3$

**Trigonometry**

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\sin^2\theta + \cos^2\theta = 1$$

# Electron arrangements of elements

**Key**

Atomic number
Symbol
Electron arrangement
Name

| Group 1 (1) | Group 2 (2) | Group 3 (13) | Group 4 (14) | Group 5 (15) | Group 6 (16) | Group 7 (17) | Group 0 (18) |
|---|---|---|---|---|---|---|---|
| 1 **H** 1 Hydrogen | | | | | | | 2 **He** 2 Helium |
| 3 **Li** 2,1 Lithium | 4 **Be** 2,2 Beryllium | 5 **B** 2,3 Boron | 6 **C** 2,4 Carbon | 7 **N** 2,5 Nitrogen | 8 **O** 2,6 Oxygen | 9 **F** 2,7 Fluorine | 10 **Ne** 2,8 Neon |
| 11 **Na** 2,8,1 Sodium | 12 **Mg** 2,8,2 Magnesium | 13 **Al** 2,8,3 Aluminium | 14 **Si** 2,8,4 Silicon | 15 **P** 2,8,5 Phosphorus | 16 **S** 2,8,6 Sulfur | 17 **Cl** 2,8,7 Chlorine | 18 **Ar** 2,8,8 Argon |
| 19 **K** 2,8,8,1 Potassium | 20 **Ca** 2,8,8,2 Calcium | 31 **Ga** 2,8,18,3 Gallium | 32 **Ge** 2,8,18,4 Germanium | 33 **As** 2,8,18,5 Arsenic | 34 **Se** 2,8,18,6 Selenium | 35 **Br** 2,8,18,7 Bromine | 36 **Kr** 2,8,18,8 Krypton |
| 37 **Rb** 2,8,18,8,1 Rubidium | 38 **Sr** 2,8,18,8,2 Strontium | 49 **In** 2,8,18, 18,3 Indium | 50 **Sn** 2,8,18, 18,4 Tin | 51 **Sb** 2,8,18, 18,5 Antimony | 52 **Te** 2,8,18, 18,6 Tellurium | 53 **I** 2,8,18, 18,7 Iodine | 54 **Xe** 2,8,18, 18,8 Xenon |
| 55 **Cs** 2,8,18,18, 8,1 Caesium | 56 **Ba** 2,8,18,18, 8,2 Barium | 81 **Tl** 2,8,18, 32,18,3 Thallium | 82 **Pb** 2,8,18, 32,18,4 Lead | 83 **Bi** 2,8,18, 32,18,5 Bismuth | 84 **Po** 2,8,18, 32,18,6 Polonium | 85 **At** 2,8,18, 32,18,7 Astatine | 86 **Rn** 2,8,18, 32,18,8 Radon |
| 87 **Fr** 2,8,18,32, 18,8,1 Francium | 88 **Ra** 2,8,18,32, 18,8,2 Radium | | | | | | |

## Transition elements

| (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
|---|---|---|---|---|---|---|---|---|---|
| 21 **Sc** 2,8,9,2 Scandium | 22 **Ti** 2,8,10,2 Titanium | 23 **V** 2,8,11,2 Vanadium | 24 **Cr** 2,8,13,1 Chromium | 25 **Mn** 2,8,13,2 Manganese | 26 **Fe** 2,8,14,2 Iron | 27 **Co** 2,8,15,2 Cobalt | 28 **Ni** 2,8,16,2 Nickel | 29 **Cu** 2,8,18,1 Copper | 30 **Zn** 2,8,18,2 Zinc |
| 39 **Y** 2,8,18,9,2 Yttrium | 40 **Zr** 2,8,18, 10,2 Zirconium | 41 **Nb** 2,8,18, 12,1 Niobium | 42 **Mo** 2,8,18,13, 1 Molybdenum | 43 **Tc** 2,8,18,13, 2 Technetium | 44 **Ru** 2,8,18,15, 1 Ruthenium | 45 **Rh** 2,8,18,16, 1 Rhodium | 46 **Pd** 2,8,18, 18,0 Palladium | 47 **Ag** 2,8,18, 18,1 Silver | 48 **Cd** 2,8,18, 18,2 Cadmium |
| 57 **La** 2,8,18,18, 9,2 Lanthanum | 72 **Hf** 2,8,18,32, 10,2 Hafnium | 73 **Ta** 2,8,18, 32,11,2 Tantalum | 74 **W** 2,8,18,32, 12,2 Tungsten | 75 **Re** 2,8,18,32, 13,2 Rhenium | 76 **Os** 2,8,18,32, 14,2 Osmium | 77 **Ir** 2,8,18,32, 15,2 Iridium | 78 **Pt** 2,8,18,32, 17,1 Platinum | 79 **Au** 2,8,18, 32,18,1 Gold | 80 **Hg** 2,8,18, 32,18,2 Mercury |
| 89 **Ac** 2,8,18,32, 18,9,2 Actinium | 104 **Rf** 2,8,18,32, 32,10,2 Rutherfordium | 105 **Db** 2,8,18,32, 32,11,2 Dubnium | 106 **Sg** 2,8,18,32, 32,12,2 Seaborgium | 107 **Bh** 2,8,18,32, 32,13,2 Bohrium | 108 **Hs** 2,8,18,32, 32,14,2 Hassium | 109 **Mt** 2,8,18,32, 32,15,2 Meitnerium | 110 **Ds** 2,8,18,32, 32,17,1 Darmstadtium | 111 **Rg** 2,8,18,32, 32,18,1 Roentgenium | 112 **Cn** 2,8,18,32, 32,18,2 Copernicium |

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Lanthanides** | 57 **La** 2,8,18, 18,9,2 Lanthanum | 58 **Ce** 2,8,18, 20,8,2 Cerium | 59 **Pr** 2,8,18,21, 8,2 Praseodymium | 60 **Nd** 2,8,18,22, 8,2 Neodymium | 61 **Pm** 2,8,18,23, 8,2 Promethium | 62 **Sm** 2,8,18,24, 8,2 Samarium | 63 **Eu** 2,8,18,25, 8,2 Europium | 64 **Gd** 2,8,18,25, 9,2 Gadolinium | 65 **Tb** 2,8,18,27, 8,2 Terbium | 66 **Dy** 2,8,18,28, 8,2 Dysprosium | 67 **Ho** 2,8,18,29, 8,2 Holmium | 68 **Er** 2,8,18,30, 8,2 Erbium | 69 **Tm** 2,8,18,31, 8,2 Thulium | 70 **Yb** 2,8,18,32, 8,2 Ytterbium | 71 **Lu** 2,8,18,32, 9,2 Lutetium |
| **Actinides** | 89 **Ac** 2,8,18,32, 18,9,2 Actinium | 90 **Th** 2,8,18,32, 18,10,2 Thorium | 91 **Pa** 2,8,18,32, 20,9,2 Protactinium | 92 **U** 2,8,18,32, 21,9,2 Uranium | 93 **Np** 2,8,18,32, 22,9,2 Neptunium | 94 **Pu** 2,8,18,32, 24,8,2 Plutonium | 95 **Am** 2,8,18,32, 25,8,2 Americium | 96 **Cm** 2,8,18,32, 25,9,2 Curium | 97 **Bk** 2,8,18,32, 27,8,2 Berkelium | 98 **Cf** 2,8,18,32, 28,8,2 Californium | 99 **Es** 2,8,18,32, 29,8,2 Einsteinium | 100 **Fm** 2,8,18,32, 30,8,2 Fermium | 101 **Md** 2,8,18,32, 31,8,2 Mendelevium | 102 **No** 2,8,18,32, 32,8,2 Nobelium | 103 **Lr** 2,8,18,32, 32,9,2 Lawrencium |

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